

Introduction to Computer Vision

CS 4243 Computer Vision & Pattern Recognition

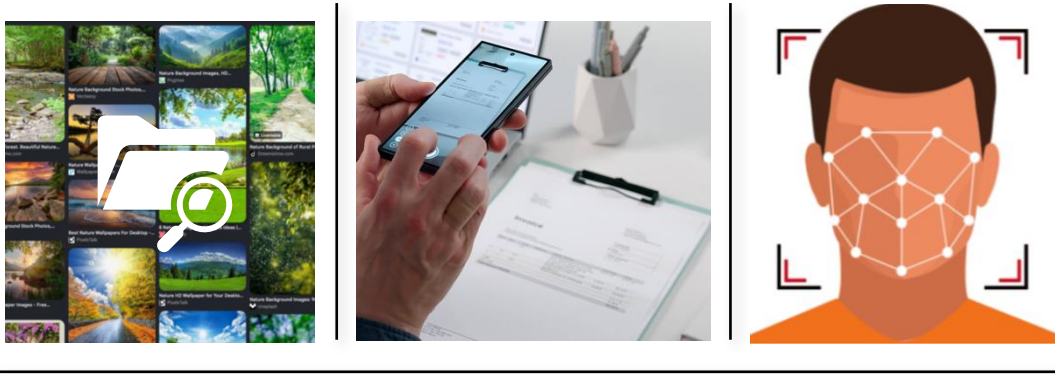
Angela Yao

Today's Lecture

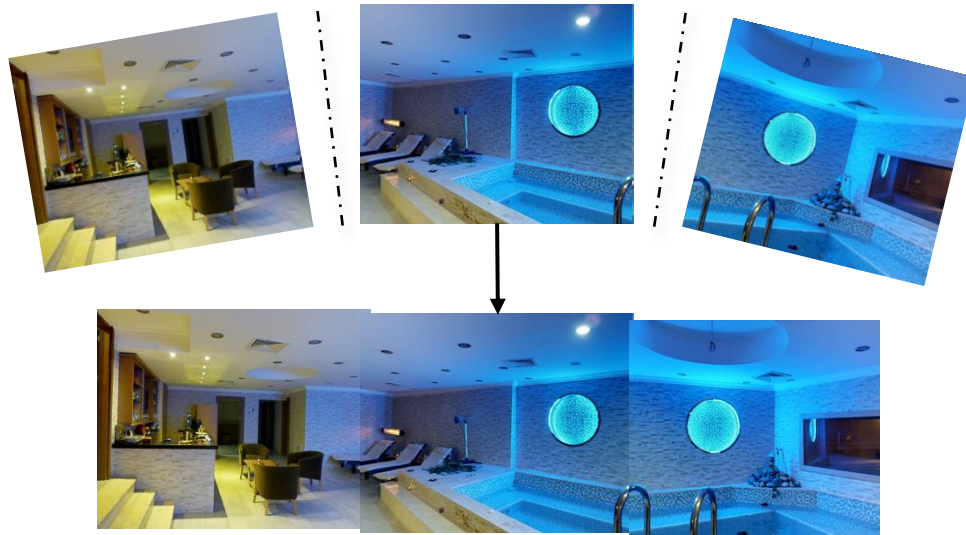
1. What is Computer Vision?
2. Why Is Computer Vision Difficult?
3. Digital Images & Colour Representations
4. Camera Pipelines & Image Compression

What is Computer Vision?

Computer Vision is Already Everywhere



- Face & fingerprint unlock
- Panorama stitching
- QR scanning
- Document scanning
- Photo search
- Visual question-answering
- Virtual backgrounds
- ...



Goals of Computer Vision



1. **Measure**: shape, depth, motion, other properties of the world from visual data

Goals of Computer Vision



1. **Measure**: shape, depth, motion, other properties of the world from visual data
2. **Recognize**: objects, people, places, activities, etc.



Goals of Computer Vision



Who threw the frisbee?

The child!



1. **Measure**: shape, depth, motion, other properties of the world from visual data
2. **Recognize**: objects, people, places, activities, etc.
3. **Understand**: infer relationships & events, describe scenes, answer questions.

Goals of Computer Vision



Plant



1. **Measure**: shape, depth, motion, other properties of the world from visual data
2. **Recognize**: objects, people, places, activities, etc.
3. **Understand**: infer relationships & events, describe scenes, answer questions.
4. **Organize**: index, mine, search, and interact with visual data

Goals of Computer Vision

Change the frisbee
to a football.

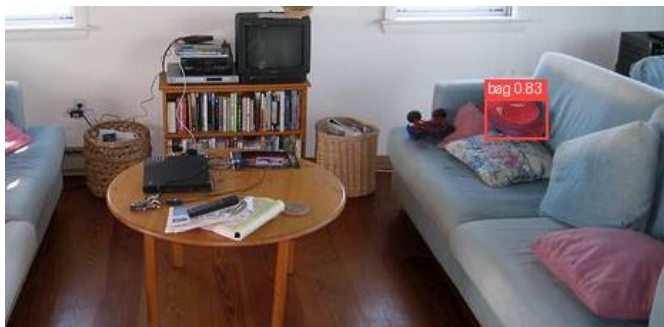


Done!



1. **Measure**: shape, depth, motion, other properties of the world from visual data
2. **Recognize**: objects, people, places, activities, etc.
3. **Understand**: infer relationships & events, describe scenes, answer questions.
4. **Organize**: index, mine, search, and interact with visual data
5. **Create**: edit or enhance existing images & videos; create new images and video

Why Computer Vision Matters



I can't find my bag!



Everyday Assistance

communication, accessibility,
convenience, entertainment



Science & Industry

Healthcare, manufacturing,
agriculture, research, ...



High-Stakes Decisions

Transportation, security,
public infrastructure, etc.

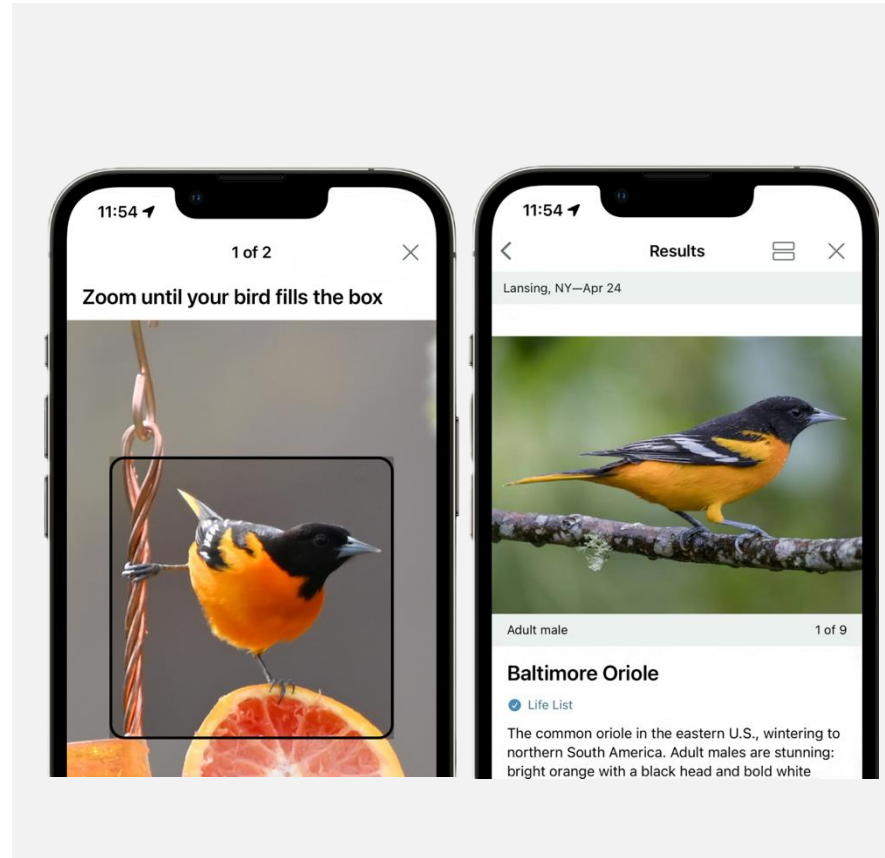
Is Computer Vision Still Hard?

Is Computer Vision Still Hard?



IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

2014



Today



Can you identify the bird?



This appears to be a
Woodchat Shrike (*Lanius
senator*)!



Why is Computer Vision Difficult?



Viewpoint variation



Illumination



Scale

Why is Computer Vision Difficult?



Intra-class variation



Motion (Source: S. Lazebnik)



Background clutter



Occlusion

Exploiting Visual Cues for Information



Visual Cues:

- visual appearance
- regions and shape
- geometry
- motion
- semantics and context

Classical vs. modern CV differ in how the features and representations are obtained: explicitly designed vs. learned from data.

Bottom Line: Vision

- Visual perception is an inherently ambiguous



- Resolving ambiguity requires assumptions and prior knowledge about the world (hand-designed or learned from the data).

Some historical Notes

1966: Papert assigns computer vision as an [undergrad summer project](#)

1980's: First wave of ANNs; shift toward geometry & mathematical rigor

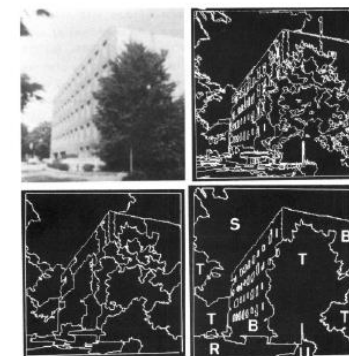
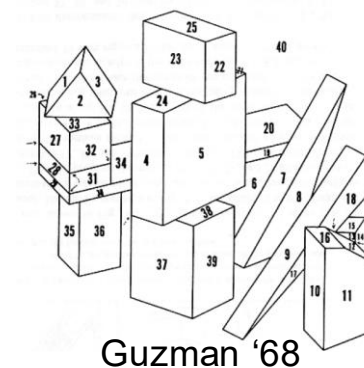
1990's: Face recognition; statistical analysis

2000's: Broader recognition; large datasets available; video processing

2010's: Deep learning with CNNs

2020's: Multimodal and foundation models, generative AI

2030's: Robot uprising?



Ohta Kanade '78

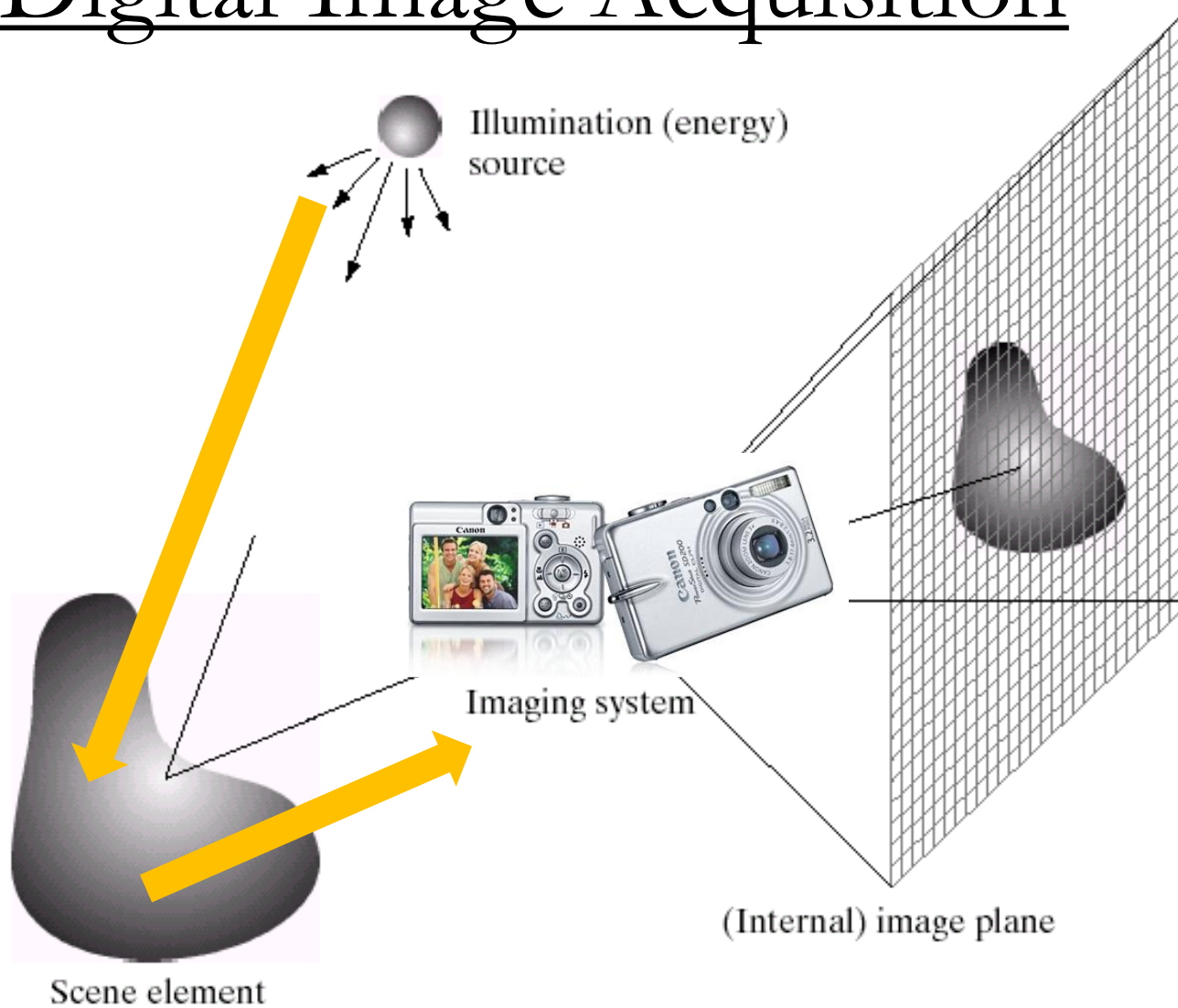


Turk and Pentland '91

Digital Images

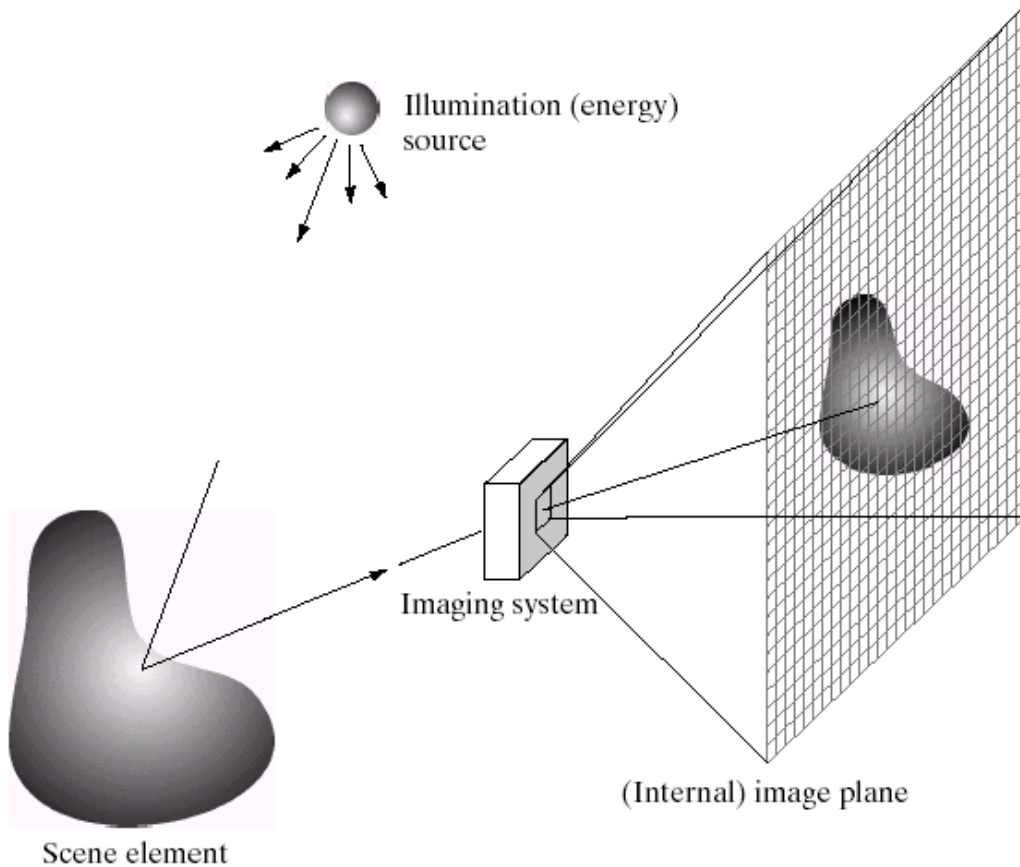
Imaging sensors, exposure time, spatial and intensity resolution

Digital Image Acquisition



1. Energy transfer from source (sun, light bulb) to scene to imaging system
2. Optics in imaging system focuses energy onto sensor (e.g. CCD/CMOS).
3. Digital sensor measures amount of energy

Simplified Image Formation Model

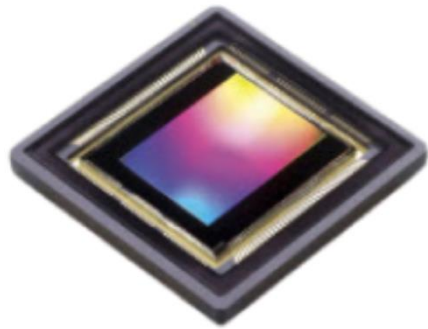


- $f(x, y)$ - 2D image function, where x, y are coordinates on the internal image plane
- Illumination component $i(x, y)$: amount of source illumination incident on the scene
- Reflectance component $r(x, y)$: amount of illumination reflected by objects in the scene

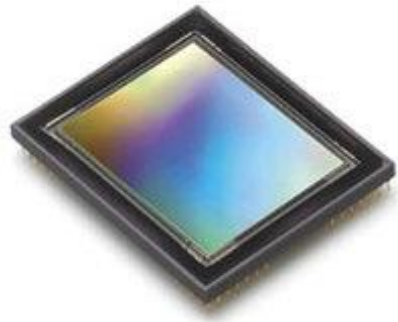
$$f(x, y) = i(x, y)r(x, y)$$

- $0 \leq i(x, y) < \infty$ - no upper bound
- $0 \leq r(x, y) \leq 1$
0 for total absorption, 1 for total reflectance

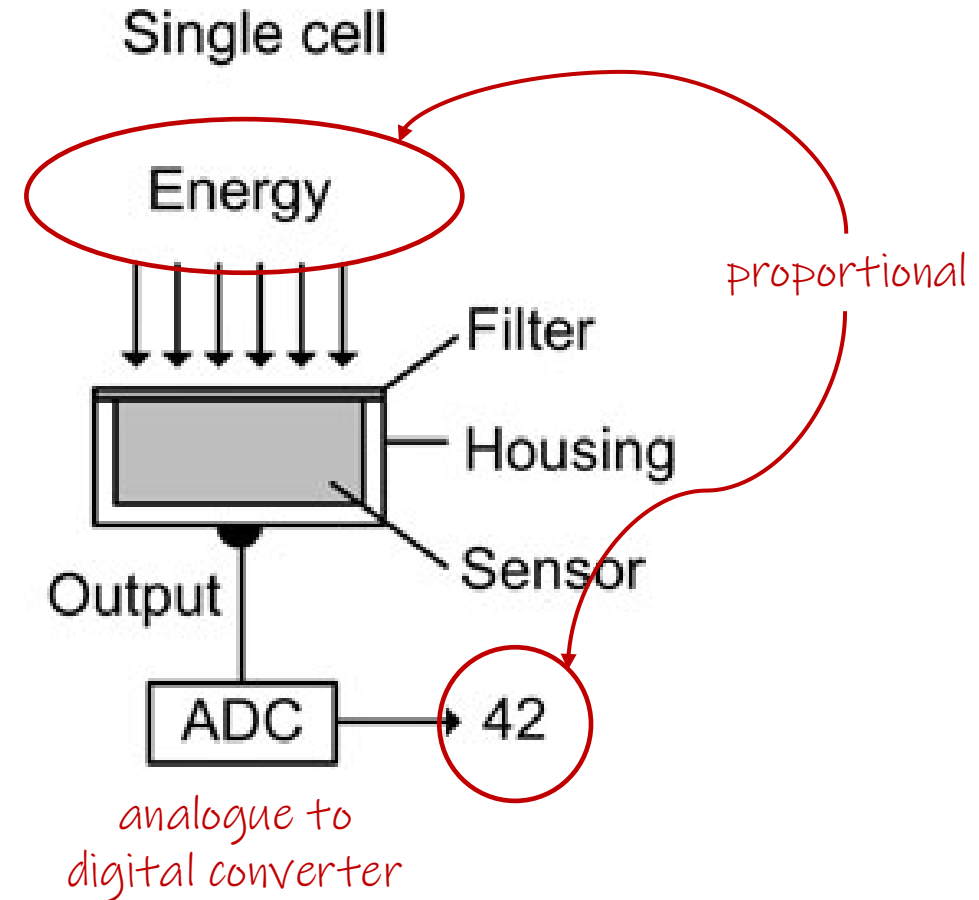
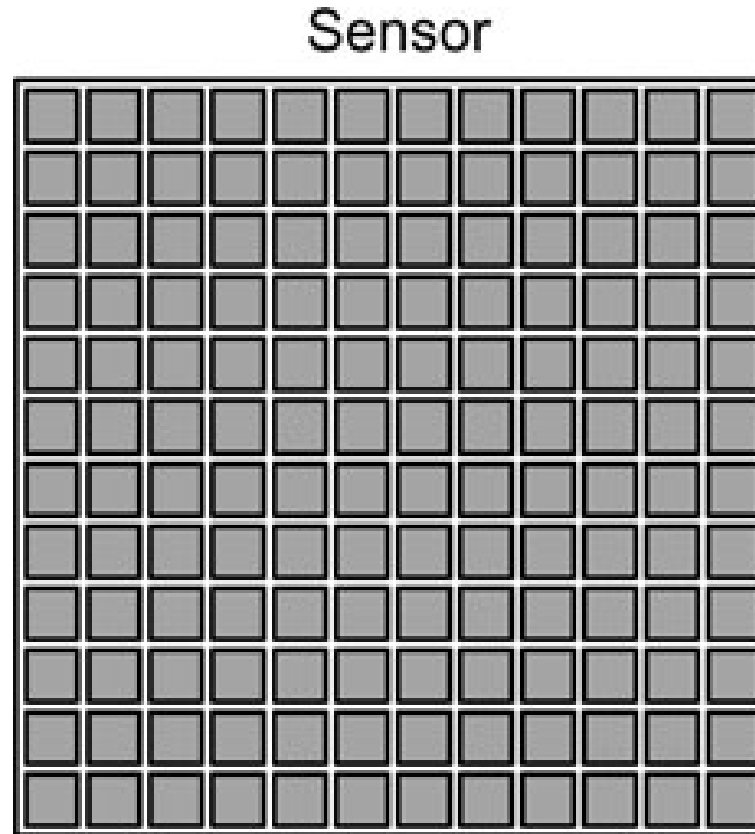
Imaging System Sensors



CMOS Sensor



CCD Sensor



Exposure Time

Exposure is the amount of time that incident light can reach the imaging sensor.



Correctly exposed



Over exposed

exposure time too long
too much light reaches sensor



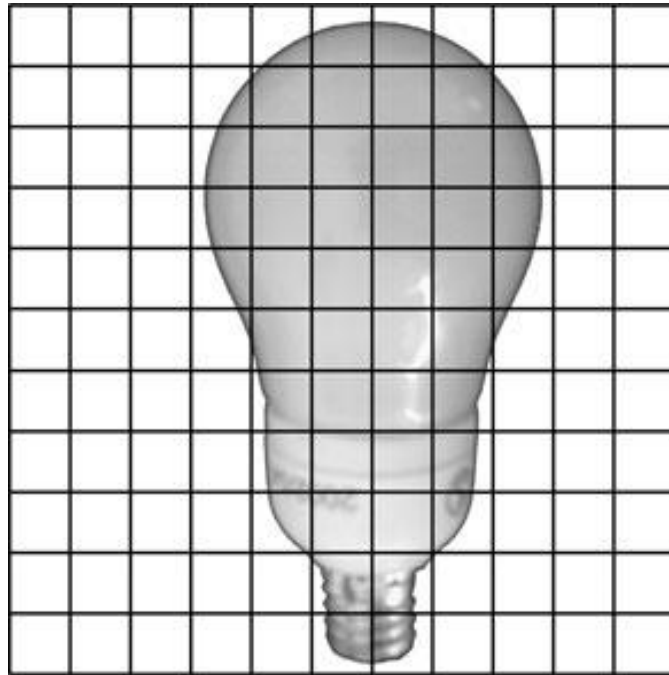
Under exposed

exposure time too short
not enough light reaches sensor

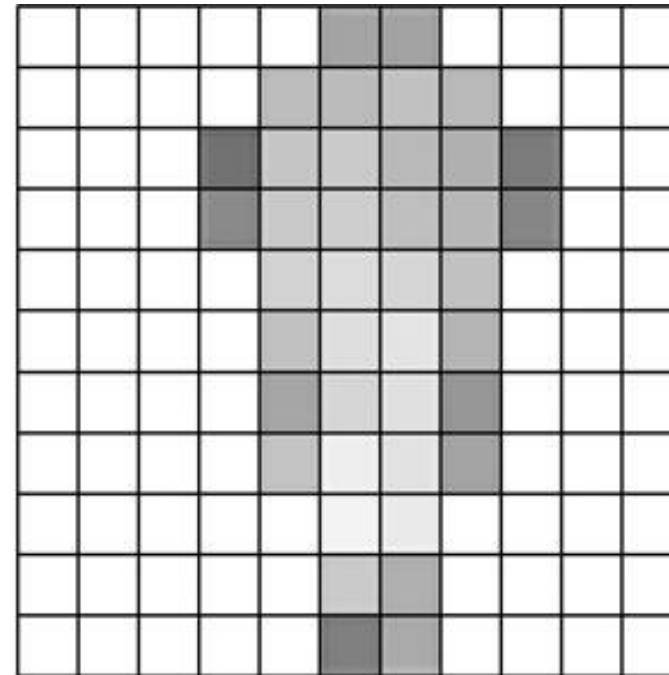


Motion blur

Image Formation



continuous projection
onto imaging sensor



digital sampling

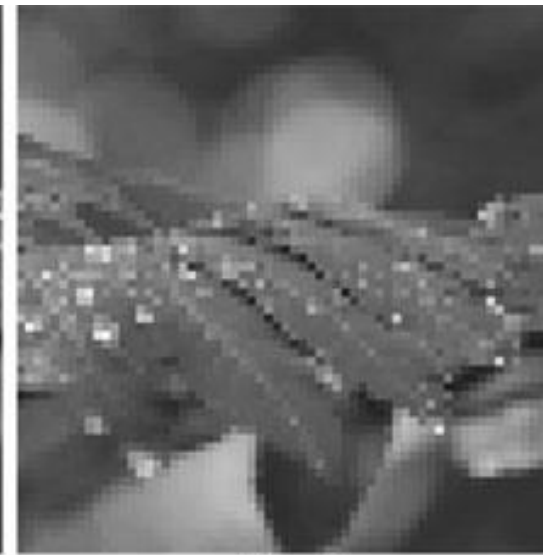
spatial resolution
intensity resolution

Resolution

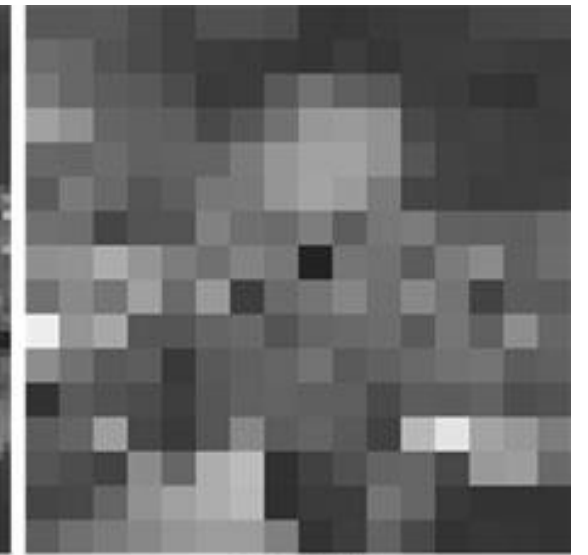
Spatial Resolution



256 x 256

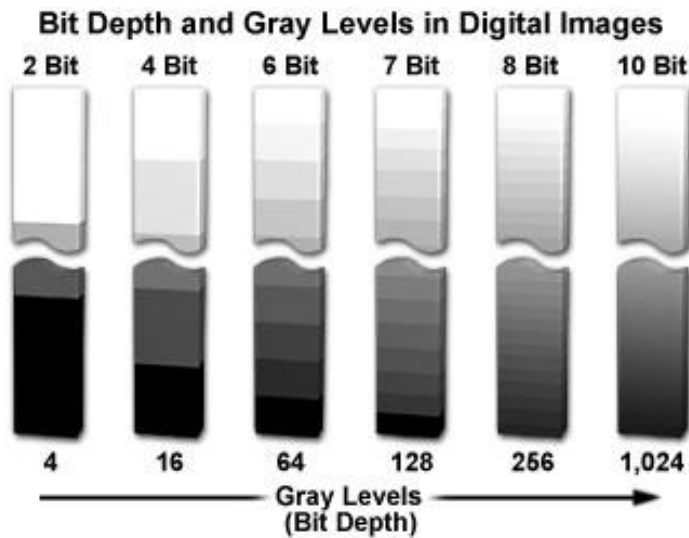


64 x 64

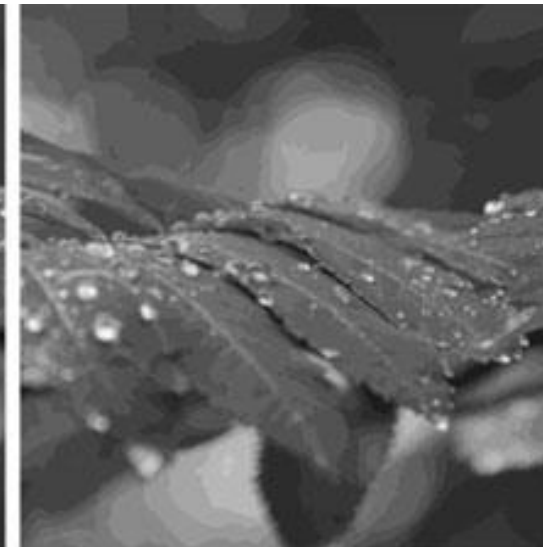


16 x 16

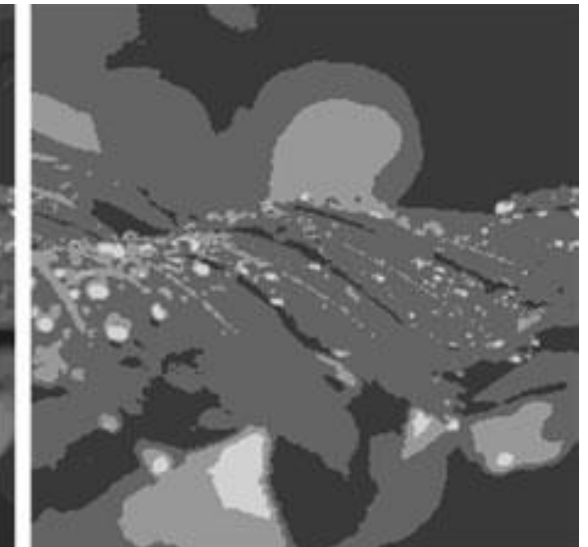
Intensity Resolution



256 gray-levels



16 gray-levels



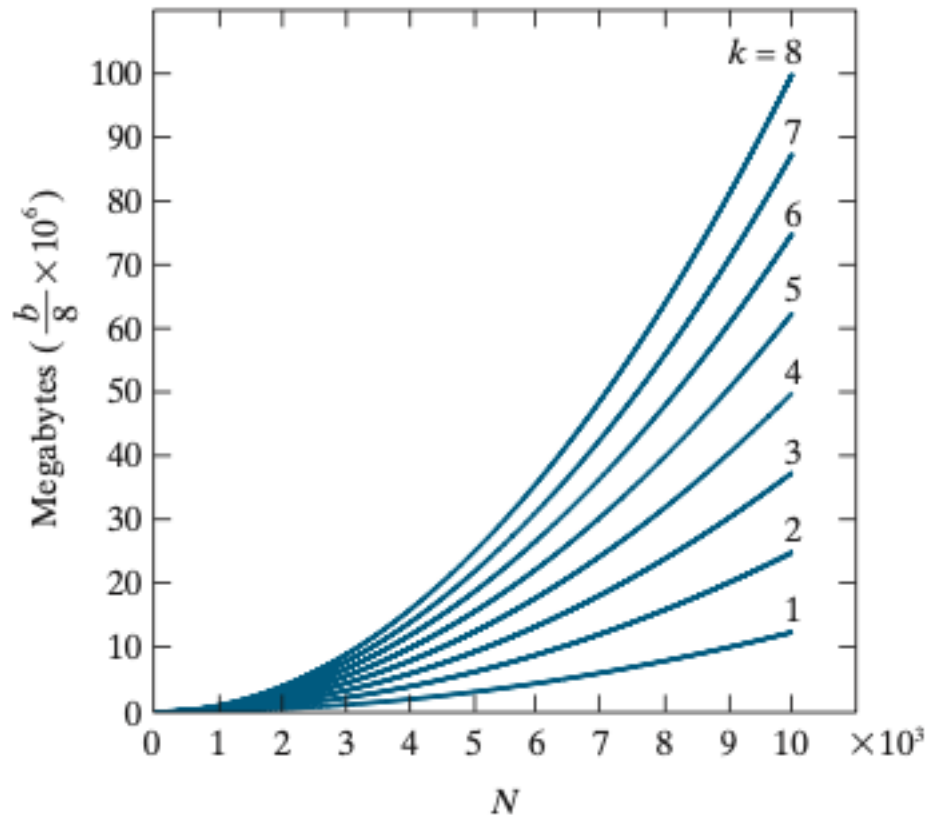
4 gray-levels

*we almost always work with
8 bits, i.e. from 0-255*

some visible false contouring

Resolution Determines File Size

Number of megabytes required to store images for various values of N and k .



- Spatial and intensity resolution determines the file size and also transmission time if we were to send images over the internet
- An image of size $M \cdot N$ with a bit-depth of k requires b bits to store, where
$$b = M \cdot N \cdot k$$
- b is the storage requirement for a raw bitmap (.bmp). Compressed formats (.png, .jpg) can reduce file sizes

Colour Representations

RGB images, Bayer patterns, colour to greyscale,
HSV colour space, colour-based applications

Colour Images



Compare

- Sky vs. building in the center
- Top of the fountain vs. clouds
- $[0,0,0]$ is black, $[255,255,255]$ is white

$[R, G, B] : [120, 30, 24]$



red channel



green channel



blue channel

Images in Python

Diagram illustrating an RGB image represented as a 3D matrix. The matrix is shown as a stack of three 10x10 grids (Red, Green, and Blue channels). A blue arrow labeled "row" points downwards on the left, and a blue arrow labeled "column" points to the right at the top. The Red channel (R) is the top grid, the Green channel (G) is the middle grid, and the Blue channel (B) is the bottom grid. The values in the grids are normalized to the range [0, 1].

row \ column	0	1	2	3	4	5	6	7	8	9
0	0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92
1	0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95
2	0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91
3	0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97
4	0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79
5	0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45
6	0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49
7	0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82
8	0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90
9	0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93

[0,255] → [0,1] for arithmetic convenience

represented as a matrix

N x M RGB image “im”

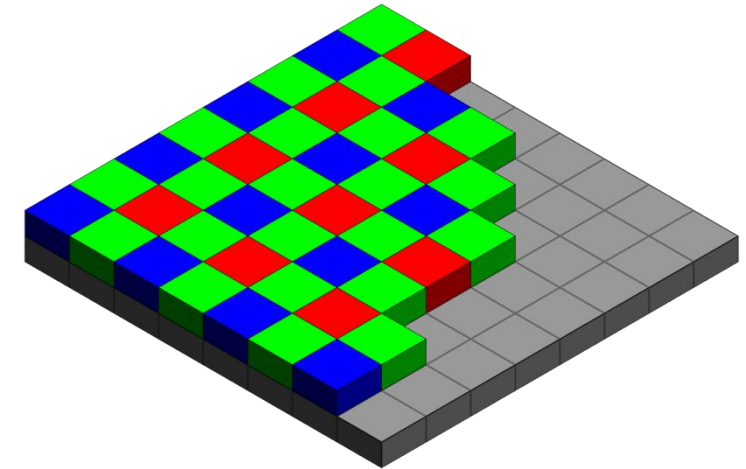
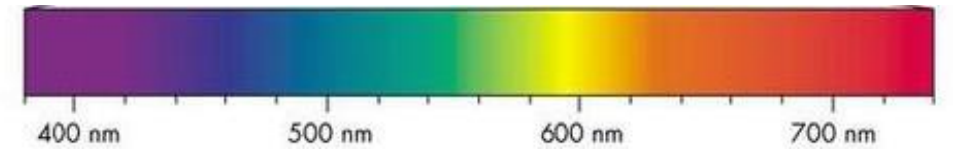
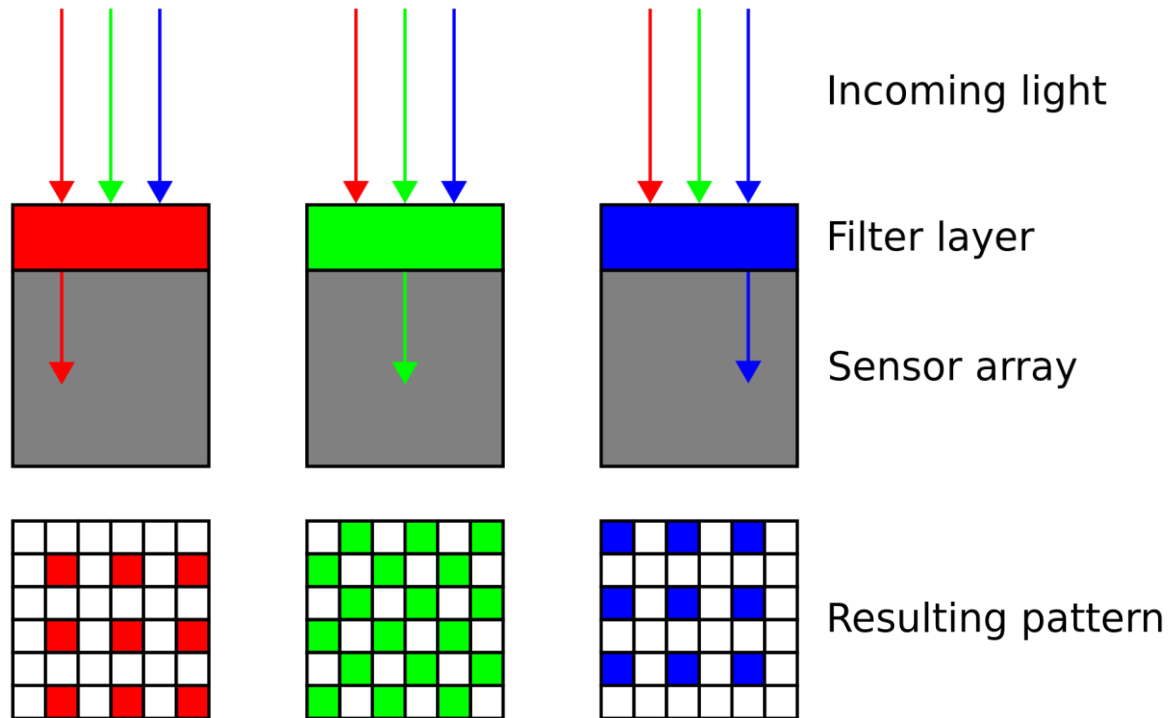
im(0,0,0) = top-left pixel value in R-channel

B im(y, x, b) = y pixels down, x pixels to right in the bth channel

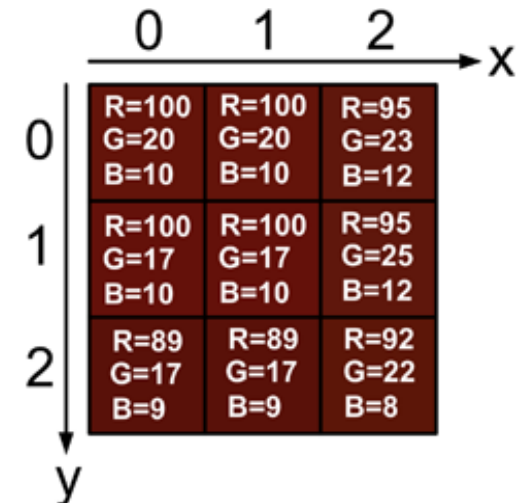
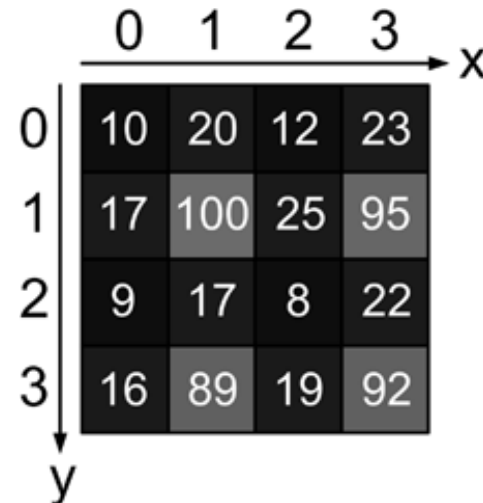
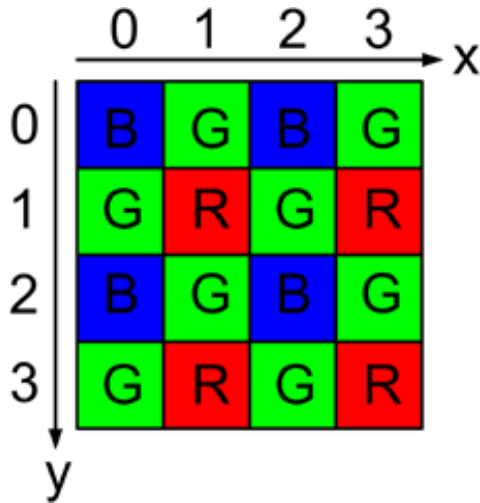
im(N-1, M-1, 2) = bottom-right pixel in B-channel

RGB is the most common order, some libraries use BGR. Always check the convention used!

Colour Capture – Bayer Filter



Demosaicing Bayer Patterns



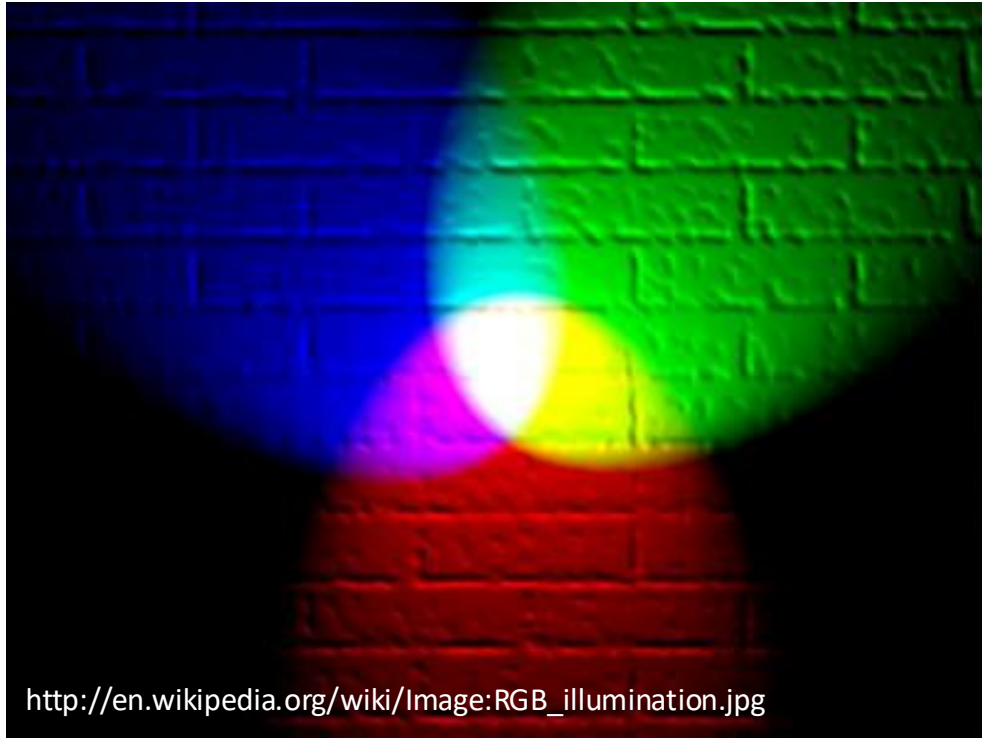
$$g(x, y) \left\{ \begin{array}{l} [R, G, B]_B = [f(x + 1, y + 1), f(x + 1, y), f(x, y)] \\ [R, G, B]_{GB} = [f(x, y + 1), f(x, y), f(x - 1, y)] \\ [R, G, B]_{GR} = [f(x + 1, y), f(x, y), f(x, y - 1)] \\ [R, G, B]_R = [f(x, y), f(x - 1, y), f(x - 1, y - 1)] \end{array} \right. \quad \left. \vphantom{\begin{array}{l} [R, G, B]_B \\ [R, G, B]_{GB} \\ [R, G, B]_{GR} \\ [R, G, B]_R \end{array}} \right\}$$

sensitive to blue

sensitive to green followed by
blue / red in row

sensitive to red

RGB Colour Model



- RGB is accepted as the “default” colour space
- It is an **additive** model used in electronic displays & photography
- Designed based on **trichromatic** human perception of colours

Colour to Greyscale

greyscale
intensity

Weight factor

$$I = W_R \cdot R + W_G \cdot G + W_B \cdot B$$

$$W_R + W_G + W_B = 1$$

keeps values
within [0, 255]

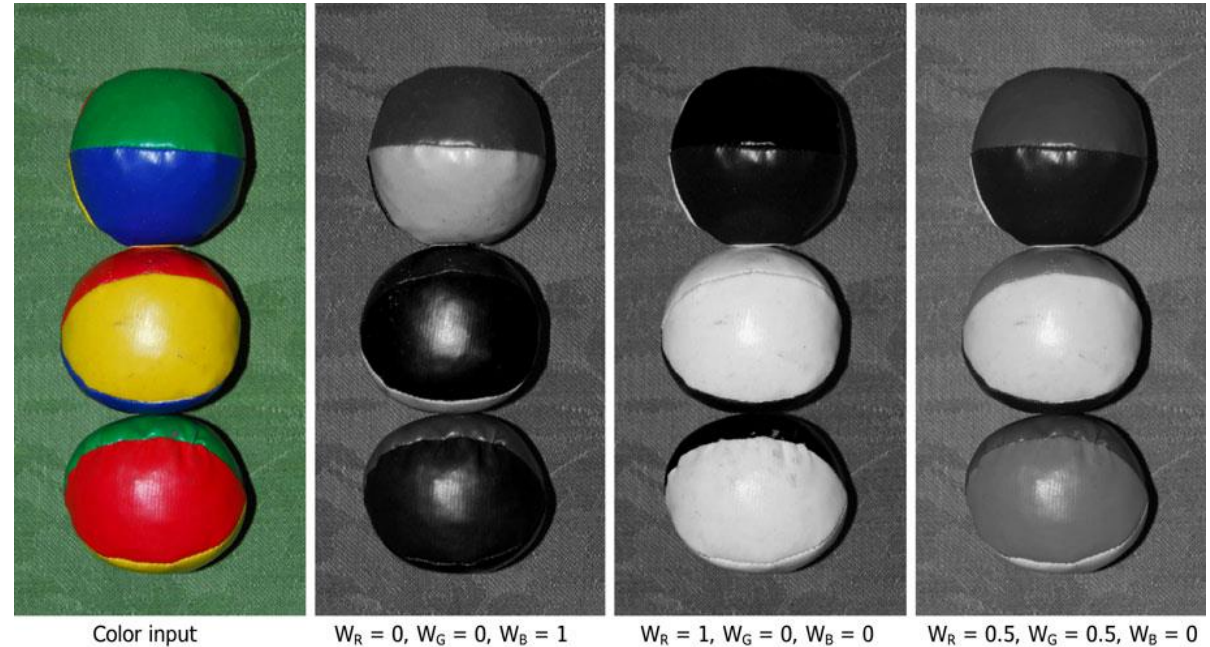
$$W_R = W_G = W_B = \frac{1}{3}$$

default, but can be adjusted
depending on application

$$W_R = 0.299, \quad W_G = 0.587, \quad W_B = 0.114$$

good compromise for
visualization purposes

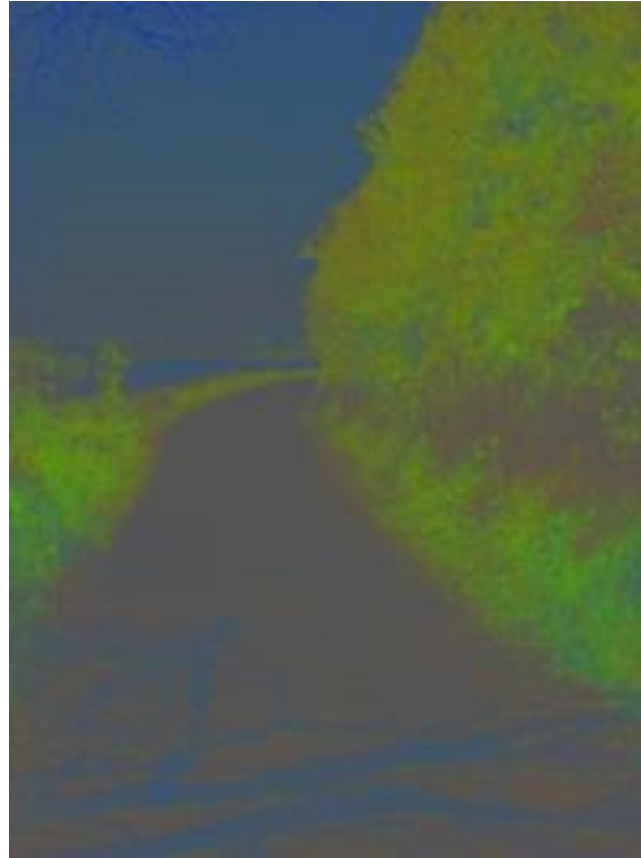
Note: Can vary depending on application



Colour and Intensity are Complementary



Original RGB
Image



Only color shown -
Constant intensity



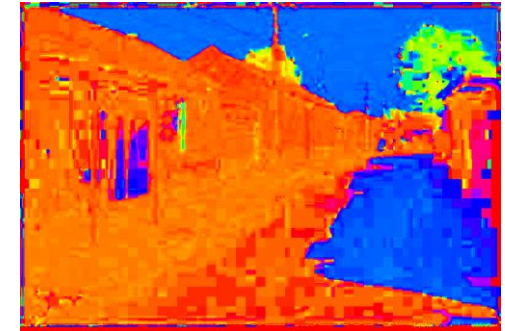
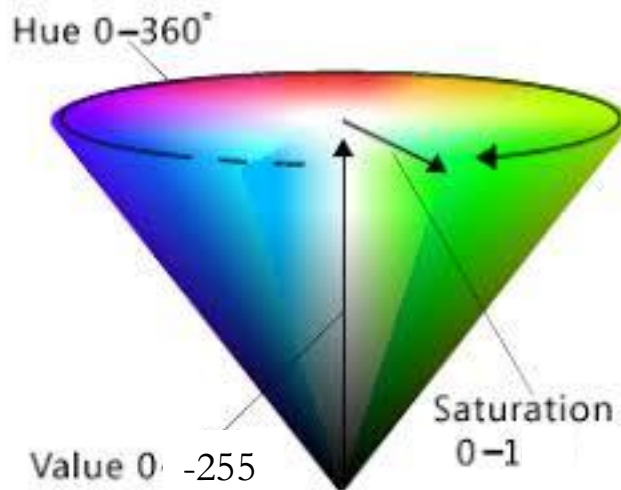
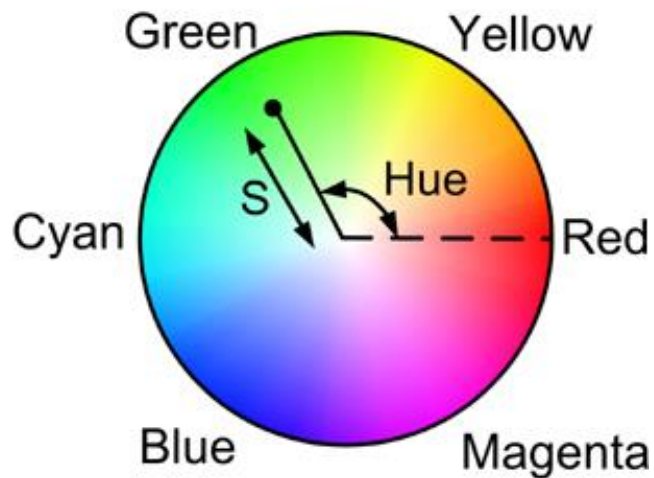
Only intensity shown
– Constant color

HSV Colour Space

Hue: “pure” colour (0 to 360)

Saturation: “purity”, mixing pure colour (1) with white light (0)

Value: achromatic mixing from black (0) to white (255)



H

(S=1,V=1)



S

(H=1,V=1)



V

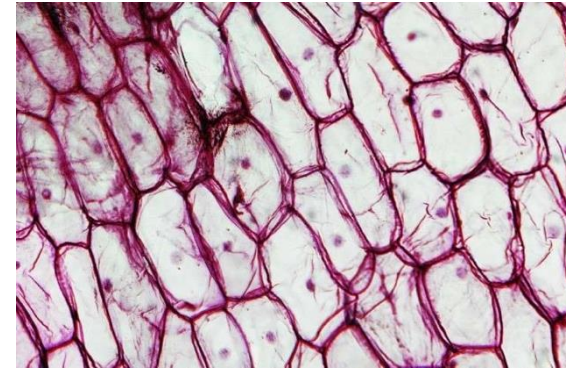
(H=1,S=0)

What is Colour Good For?

Colour based analysis: Ripeness



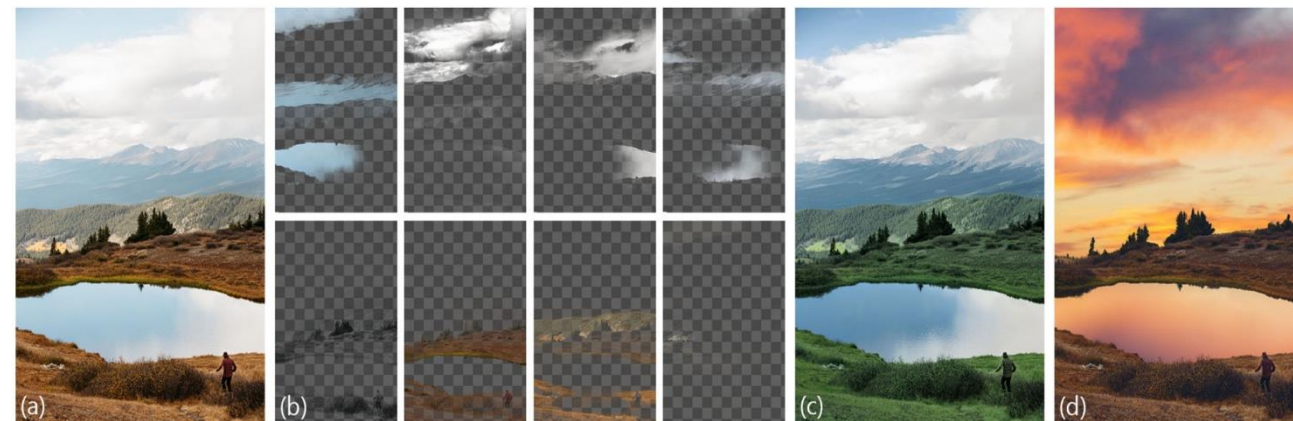
Colour based analysis: Cell Dyes



Colour based selection: Replace Background



Colour based selection: Photo-editing



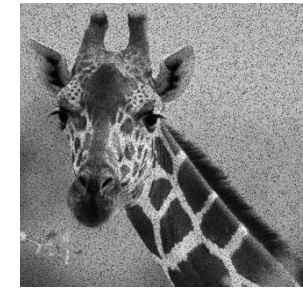
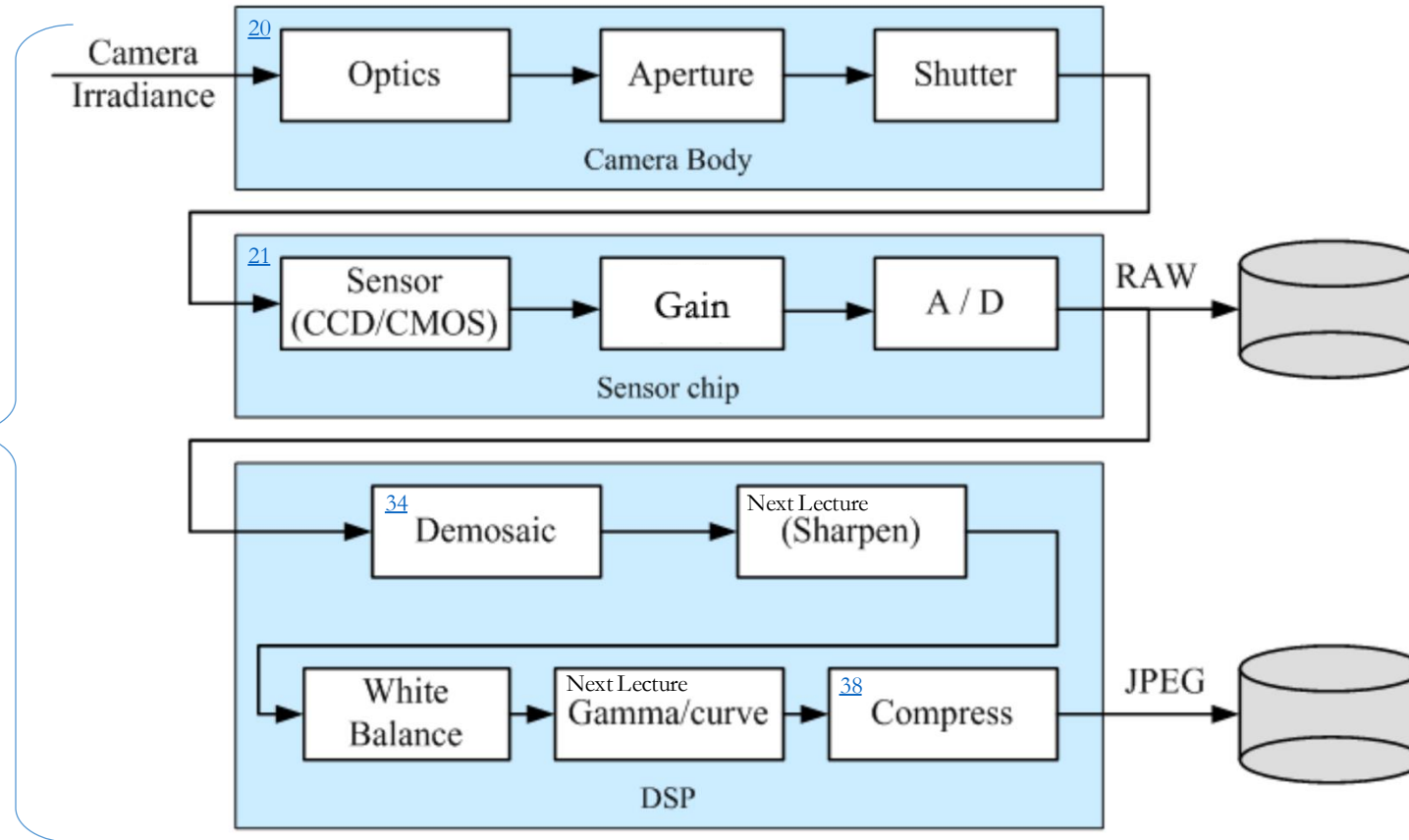
Camera Pipeline

Modern camera pipeline, image formats,
JPEG compression, compression artifacts

Modern Camera Pipeline

Aperture controls the size of the opening inside the lens, while **shutter speed** controls how long the camera's sensor is exposed to light. Together, they help determine how much light is captured.

NOTE: these are the major blocks, but how each are implemented is proprietary to the manufacturer



White balance adjusts the relative strengths of the red, green, and blue channels so neutral colors appear neutral

Modern Camera Pipeline ++

Modern camera systems use learned models both to control image capture and to replace, combine, or extend traditional image-processing stages.

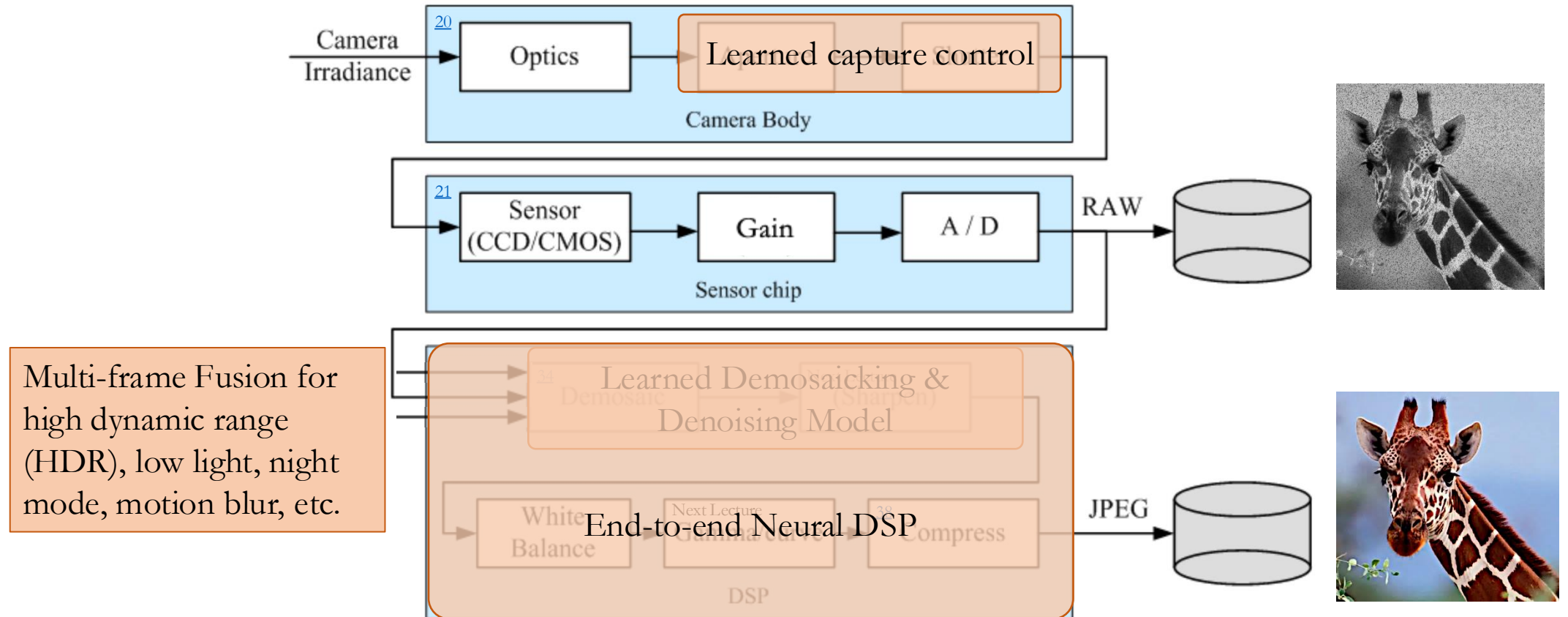


Image Formats



- Bitmap
- Stores pixel values almost directly as a rasterized array
- simple to read out as image, but normally produces very large files
- Good when simplicity and individual pixel access matters more than storage size

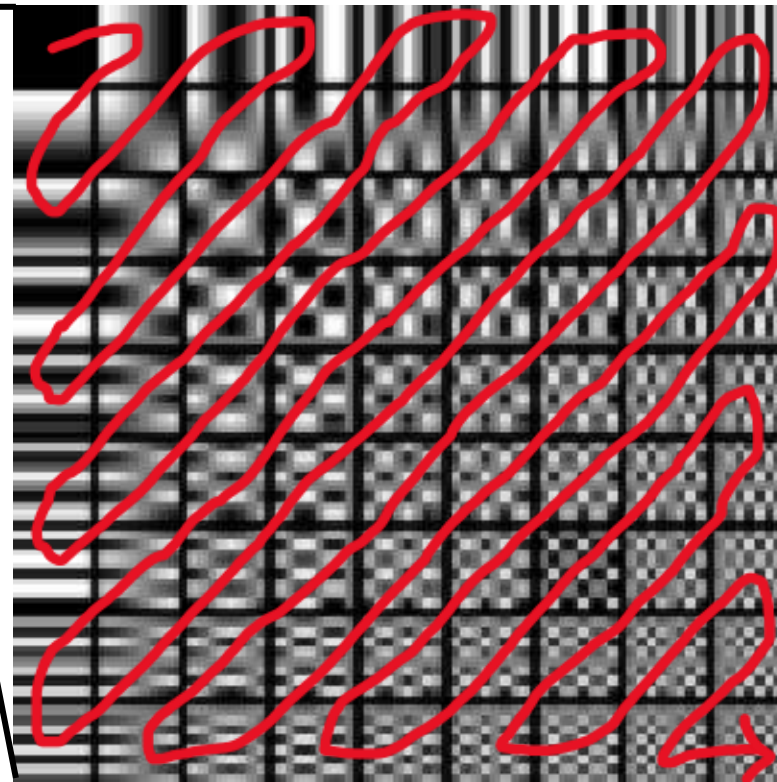
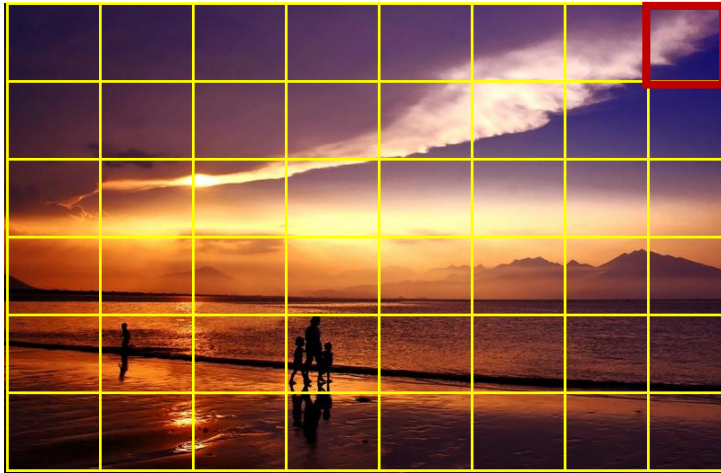


- Portable Network Graphic
- Structured storage, predicting pixels based on neighbours
Reconstructs the original pixel value exactly (lossless)
- Effective for diagrams, text, screenshots, graphics with repeated regions



- Joint Photographic (E)xperts Group
- Represents 8x8 image blocks with quantized spatial frequency coefficients
- Reconstructed pixels are an approximation (lossy)
- Effective for photos, natural “real-world” images

JPEG Compression



**Discrete Cosine Transform
(DCT) basis patterns**

- Each 8×8 image block is represented by 64 DCT coefficients
- The DCT coefficients are the weights assigned to the DCT basis patterns; their weighted sum reconstructs the image block.
- More coefficients set to zero means higher compression, but also more detail loss
- Zig-zag ordering groups the zeros together so they can be encoded efficiently.

JPEG Compression Artifacts



ringing and blocking
of wave edge



colour banding
of the sky

- **General blurring:** Loss of fine details and sharp edges as high-frequency content is removed
- **Blocking:** Visible square block boundaries caused by block-based compression
- **Ringings:** Ripple-like halos around sharp edges due to high-frequency information loss.
- **Colour Banding:** Colour subsampling reduces colour precision and turns smooth colour gradients into visible bands (a form of intensity

Summary

What computer vision does

- Measure, recognize, understand, organize and create
- Supports everyday applications, science, industry and high-stakes decisions

Why vision is difficult

- Images vary with viewpoint, scale, illumination, motion, clutter and occlusion
- Visual evidence is inherently ambiguous
- Resolving ambiguity requires visual cues and prior knowledge

Digital Images

- Sensors measure light reflected from the scene
- Exposure affects how much light is captured
- Spatial sampling determines pixel resolution
- Bit depth determines intensity resolution and affects storage

Camera Pipelines

- Bayer sensors capture incomplete colour samples; demosaicing reconstructs RGB
- RGB, greyscale and HSV expose different information
- BMP, PNG and JPEG make different storage-quality trade-offs

Take-Home Challenge: Can You Fool AI?

Task:

- Take two photos of the same everyday object or scene, one “easy” and one “hard”
- Easy: show the object or scene clearly from a familiar viewpoint.
- Challenge: Make it difficult for an AI to recognize, but still recognizable to a human.

Consider challenge factors such as unusual viewpoint, partial occlusion, tiny scale, background clutter, poor illumination, motion blur, etc.

** Please do not do post-processing of your photo beyond what the camera gives as output.

Show your challenge photo to your favourite multi-modal chatbot (ChatGPT, Claude, Gemini, ...) and ask it to recognize what is in the photo.

Submit one .pptx slide by Sunday Aug. 16 22:00:

- Both photos
- The techniques you used and the challenge factor you tried to incorporate
- The AI’s response (share a screenshot)
- One sentence explaining why a human might still succeed

Extra Readings

- [Szeliski](#) 1.1, 1.2 What is computer vision & A brief history
- [Szeliski](#) 2.3, 2.3.2 The digital camera & colour
- [Moeslund](#) Chapter 3 Color Images
- [Torralba](#) Chapters 1-4: these give a beautiful treatment of the broader aspects of computer vision, its challenges, history, etc.
- [OpenCV](#) Documentation on Color Space Conversions: A practical reference for RGB, greyscale, HSV, Bayer, and implementation-dependent ranges

AI-Generated FAQs

B1. Computer Vision and Perception

Q1. Is computer vision simply object recognition?

No. Recognition is one goal. Computer vision also measures geometry and motion, interprets relationships and events, organizes visual collections, supports decisions and actions, and creates or edits visual content.

Q2. If foundation and multimodal models are so capable, is computer vision still difficult?

Yes. Modern models handle many variations much better than earlier systems, but visual measurements remain incomplete and ambiguous. Occlusion, unfamiliar conditions, small or rare objects, distribution shifts, and insufficient context can still cause failure. High average performance also does not guarantee reliable behaviour in every setting.

Q3. Do modern learned systems still use visual cues?

Yes. The distinction is mainly how those cues are represented and obtained. Classical methods often specified edges, corners, motion, geometry, or colour measurements explicitly. Learned systems infer useful representations from data, but their predictions still depend on information in the image and on learned prior knowledge.

Q4. Why include generation as a goal of computer vision?

Modern visual systems do not only analyze images. They can restore, edit, synthesize, or generate visual content, often conditioned on images, text, geometry, or other measurements. Generation also relies on learned representations of visual appearance and structure.

B2. Image Capture and Sensing

Q5. What is the difference between physical light intensity, image intensity, and intensity resolution?

Physical irradiance is a continuous quantity describing light arriving at the sensor. An image intensity value is the numerical value used to represent a measurement after sensor response and processing. For an 8-bit image it is often stored in the range 0-255, but that range is a representation choice, not a physical definition of light. Intensity resolution, or bit depth, is the number of numerical levels available: (2^b) levels for (b) bits.

Q6. Does the illumination term refer to one light source?

Not necessarily. In a simplified illumination-reflectance model, illumination represents the combined incident lighting at a surface point. It may include the sun, artificial lights, skylight, reflections from other surfaces, and spatial effects such as shadows. The product model is an approximation used for reasoning.

Q7. What is camera gain, and is it the same as collecting more light?

Gain changes the amplification or numerical scaling applied to the sensor signal. Increasing exposure time or aperture collects more photons; increasing gain does not. Higher gain can make the output brighter, but it also changes the visibility of noise and reduces the available headroom before values saturate. Cameras may use both analog gain before digitization and digital gain afterward.

Q8. How do aperture, shutter time, and gain differ?

- Aperture controls the rate at which light enters and also affects depth of field; many phone cameras have a fixed aperture.
- Shutter time controls how long light is collected; longer exposures gather more light but increase motion blur.
- Gain amplifies the captured sensor signal; it can brighten the result but cannot recover photons that were never measured.

Automatic or learned capture control balances these settings using scene brightness, motion, and camera constraints.

Q9. What is the difference between RAW and JPEG?

RAW contains minimally processed sensor measurements, usually with higher bit depth and a colour-filter mosaic. JPEG contains a rendered colour image after operations such as demosaicing, white balance, tone mapping, sharpening, and lossy compression. A RAW file is not a directly viewable ground-truth image; it still requires interpretation and processing.

B3. Sampling, Quantization, and Resolution

Q10. Why are many images stored using 8 bits per channel? Must computer vision use 8-bit images?

No. Eight bits per channel is a common storage and display convention because it provides 256 code values per channel with modest storage. Cameras commonly capture RAW data at higher precision, and vision pipelines frequently convert images to floating-point values for computation. The appropriate precision depends on capture, display, processing, and storage requirements.

AI-Generated FAQs

Q11. Are spatial resolution and pixel count the same?

Pixel dimensions describe the number of spatial samples in the stored image, so they are one component of spatial resolution. Effective resolved detail also depends on optics, sensor size and pixel pitch, focus, motion, noise, demosaicing, compression, and subsequent processing. More pixels do not automatically mean more genuine scene detail.

Q12. What is the difference between spatial resolution and intensity resolution?

Spatial resolution concerns how finely the image plane is sampled: the number and spacing of pixels. Intensity resolution concerns how finely each pixel value is quantized: the number of levels available per channel. Increasing one does not automatically increase the other.

B4. Colour and Demosaicing

Q13. How can a sensor measure colour if each photosite produces only one electrical measurement?

A colour filter changes the photosite's spectral sensitivity, so that it responds mainly to a selected range of wavelengths. In a Bayer array, neighbouring photosites have red-, green-, or blue-selective filters. The filters and sensor responses overlap and are not perfectly exclusive; an unmeasured colour component is unknown rather than physically guaranteed to be zero.

Q14. When displaying an individual RGB channel, are the other channels set to zero?

The stored RGB image still contains all three channels. A channel can be visualized as a greyscale image containing its values, or as a colour image with the other displayed channels set to zero. This display choice is different from Bayer capture, where each photosite initially provides only one filtered measurement.

Q15. Does demosaicing change the image dimensions?

It preserves the spatial dimensions but changes the representation. An $(H \times W)$ Bayer mosaic with one recorded value at each photosite becomes an $(H \times W \times 3)$ RGB array by estimating the two missing colour components at each position. Demosaicing does not create three independent sensor measurements at every pixel; two components are inferred.

Q16. Why are there two kinds of green location in a Bayer pattern?

A Bayer pattern contains green samples in both red-containing and blue-containing rows. They belong to the same output green channel but occupy different positions in the sampling lattice. Demosaicing uses the local arrangement to determine which neighbouring red, green, and blue measurements are available for interpolation.

Q17. Are greyscale-conversion weights fixed?

Libraries normally use a defined set of coefficients associated with a particular colour convention, but other mappings are possible. The coefficients reflect the greater contribution of green than blue to perceived luminance. Equal weighting is simple but does not approximate perceived brightness as well. The weights are normally chosen by the representation or application, not adjusted independently for every photograph.

Q18. If greyscale often preserves scene structure, why retain colour?

Intensity variations often reveal boundaries, shape, texture, and shading, but colour can distinguish regions with similar intensity and can be essential for recognition, material interpretation, biological or medical analysis, and semantic conventions such as traffic signals. Neither representation is universally more informative; usefulness is task-dependent.

Q19. Why use HSV instead of RGB?

HSV rearranges colour information into hue, saturation, and value, which can make interactive colour selection and some colour-based rules easier to express. It does not create new information and is not automatically robust to lighting. Hue is unstable when saturation is low, and software libraries may use different numerical ranges.

B5. File Formats and JPEG

Q20. What is the fundamental difference between BMP, PNG, and JPEG?

After decoding, all three produce raster pixel arrays. BMP commonly stores pixel values almost directly, PNG compresses them without changing the decoded values, and JPEG transforms and quantizes image information to achieve greater compression at the cost of irreversible error.

Q21. What does lossless compression mean?

Lossless compression allows the exact original symbol sequence or pixel values to be reconstructed. It does not mean that the file is uncompressed. PNG is compressed but lossless; a typical JPEG photo is compressed and lossy.

AI-Generated FAQs

Q22. What are the DCT coefficients in JPEG?

The 64 DCT basis patterns form a coordinate system for an (8×8) block. Each coefficient is the weight assigned to one basis pattern. Adding the 64 weighted patterns reconstructs the block before quantization. Low frequency coefficients describe slowly varying structure; high-frequency coefficients describe rapid changes and fine detail.

Q23. Does the zig-zag scan remove high-frequency information?

No. Quantization causes information loss and often turns many high-frequency coefficients into zeros. The zig-zag scan then orders the quantized coefficients roughly from low to high frequency, placing long runs of zeros together so they can be encoded efficiently.

Q24. What does a JPEG quality value such as 75% mean?

It is better written as an encoder setting such as $(Q=75)$, not 75%. It is not the percentage of coefficients, information, visual quality, or original file size retained. Encoders map the setting to quantization tables, and different implementations can produce different results at the same numerical setting.

Q25. Can JPEG compression affect computer-vision results?

Yes. Strong compression can alter edges, texture, colour, and fine detail through blocking, ringing, and quantization. Mild compression may have little practical effect for some tasks, while small-object recognition, measurement, scientific imaging, or repeated recompression can be more sensitive.

B6. Modern Camera Pipelines

Q26. Is computational photography merely post-processing?

Not necessarily. Capture-time computational photography can select exposure settings, acquire multiple RAW frames, and combine their measurements before the final photograph exists. Post-capture editing starts from an already formed image. Some operations can appear in either setting, so the timing and available inputs matter.

Q27. Are low-light multi-frame fusion and HDR the same?

They use related machinery but emphasize different problems. HDR seeks to preserve detail across bright and dark regions, while low-light fusion primarily reduces noise and recovers detail from weak measurements. Modern night pipelines may perform both simultaneously.

Q28. What can a learned reconstruction model replace?

A narrowly scoped model may replace demosaicing or denoising, or learn both jointly. A broader neural ISP can learn a RAW-to-RGB mapping that also incorporates white balance, colour correction, tone mapping, sharpening, and other rendering operations. The exact scope depends on the model and training target.

Q29. Has deep learning replaced the complete camera pipeline?

Not generally. Modern cameras are hybrid systems. Physical optics and sensors remain essential; conventional algorithms remain efficient, predictable, and controllable; learned models are used where data-driven estimation or joint optimization is advantageous. Commercial implementations also vary and are often proprietary.